

Is this the best drug for me?

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Objective: Choose the best drug based on its efficacy, safety and cost-effectiveness

Knowledge:

- Understanding the strengths and limitations of clinical evidence.
- Distinguishing between efficacy and effectiveness of a treatment.
- Understanding what are absolute and relative risk reductions and odds ratios.
- Grasping the difference between statistical significance and clinical significance.
- Appreciating that individuals or subgroups may experience a different benefit-risk profile (personalised medicine).

Skills: Interpreting clinical trial, pharmacoepidemiological and pharmacoeconomic data.

Attitude: Recognising that the quality of prescriptions can be improved by considering a drug's efficacy, safety and cost-effectiveness.

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The most important properties of a drug

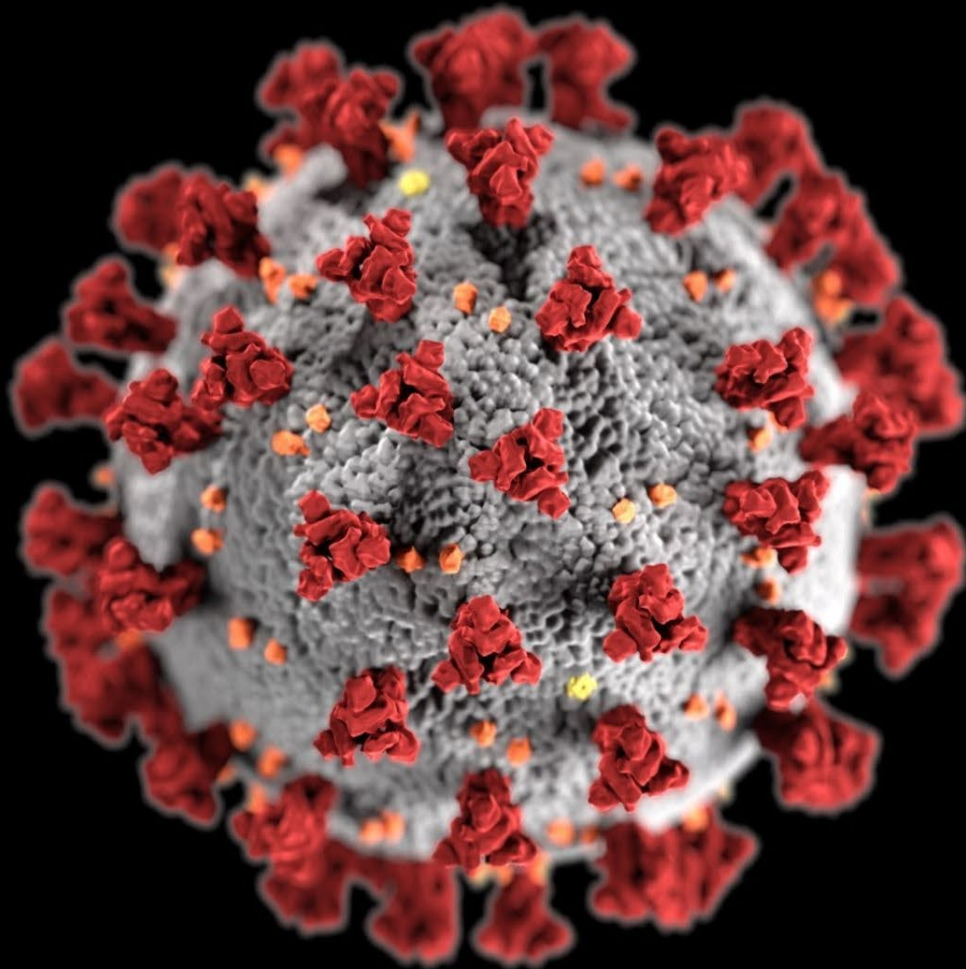
- Efficacy
- Quality & Safety
- Tolerability
- Cost-effectiveness



Why do we need new drugs?

- New conditions
- New pathogens
- New understanding of disease mechanisms
(e.g. new receptor, enzyme or pathway identified)
- New mutations

SARS-CoV-2



Why do we need new drugs?

- More choices
- To differ from generics
- Resistance
- More specificity

Why do we need new drugs?

- Better efficacy
- Better safety
- Better tolerability
- Better compliance

The evaluation of drugs



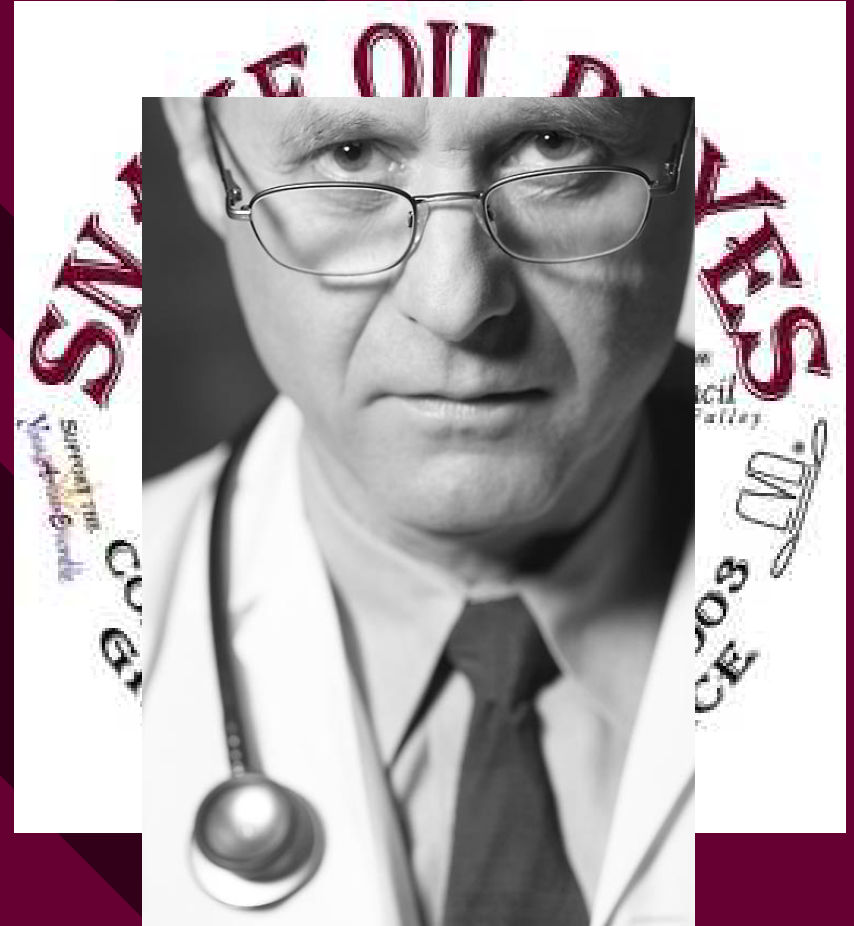
What is a clinical trial?



Assessment of a method of treatment
in a planned experiment
involving patients

Bad old days

- Traditional recipe used by generations
- Anecdotes
- Clinical experience of a lifetime
- Expert opinion - the man in the white coat recommends



Phases of clinical trials

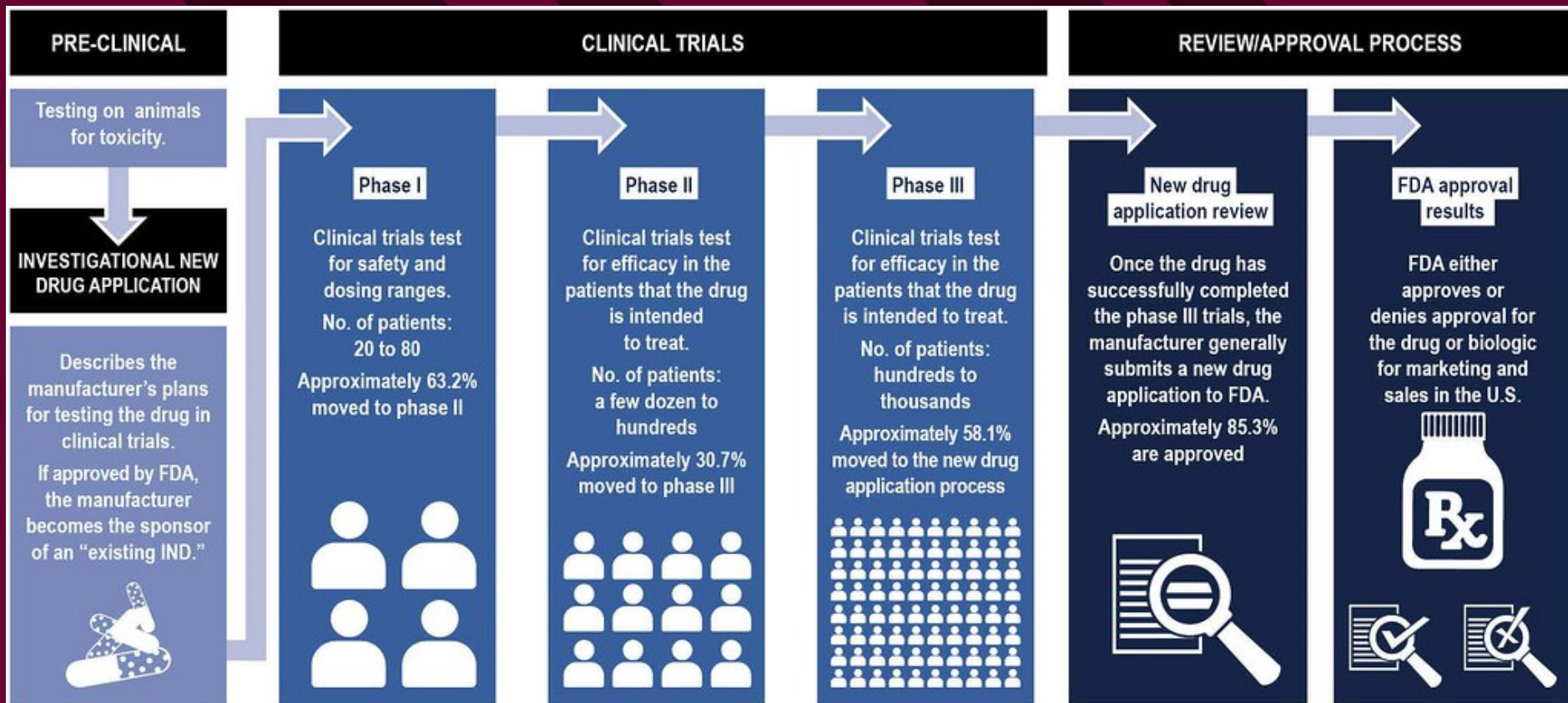
In Phase I trials, researchers test an experimental drug or treatment in a small group of people (20-80) for the first time to evaluate its safety, determine a safe dosage range, and identify side effects.

In Phase II trials, the experimental study drug or treatment is given to a larger group of people (100-300) to see if it is effective and to further evaluate its safety.

In Phase III trials, the experimental study drug or treatment is given to large groups of people (1,000-3,000) to confirm its effectiveness, monitor side effects, compare it to commonly used treatments, and collect information that will allow the experimental drug or treatment to be used safely.

In Phase IV trials, post marketing studies delineate additional information including the drug's risks, benefits, and optimal use.

How drugs are developed and approved



Source: GAO analysis of FDA data and a 2016 collaborative study by Biotechnology Innovation Organization, Biomedtracker, and Amplion.^a | GAO-17-564

HKU Phase 1 Clinical Trials Centre

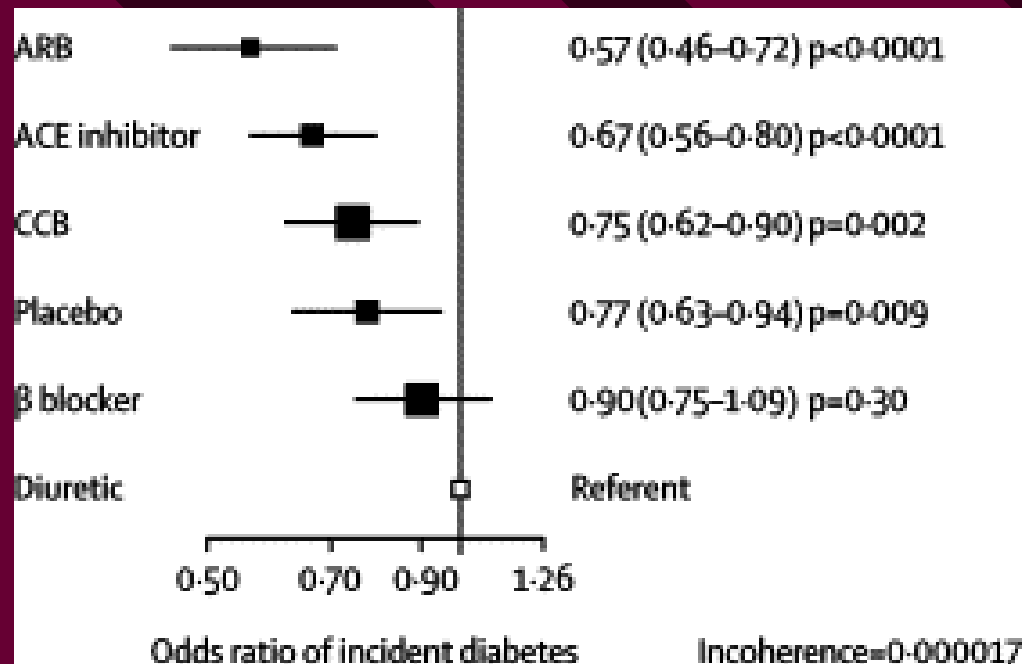


Design

- There are many trial designs
- Best trials are:
 - Prospective
 - Controlled
 - Randomised
 - Blinded

Why controlled?

- Need to know how good the new treatment is compared to no treatment (placebo) and current standard or best treatment



Why randomised?

- To even out the biases (a process at any stage of inference tending to produce results that depart systematically from true values)
- It facilitates blinding
- It allows the null hypothesis that any difference between treatment groups are due to chance

Why blinded?

Knowledge of assignment influences the
subjects' response
and the investigator's conduct and evaluation

Blind tasting



Blind audition



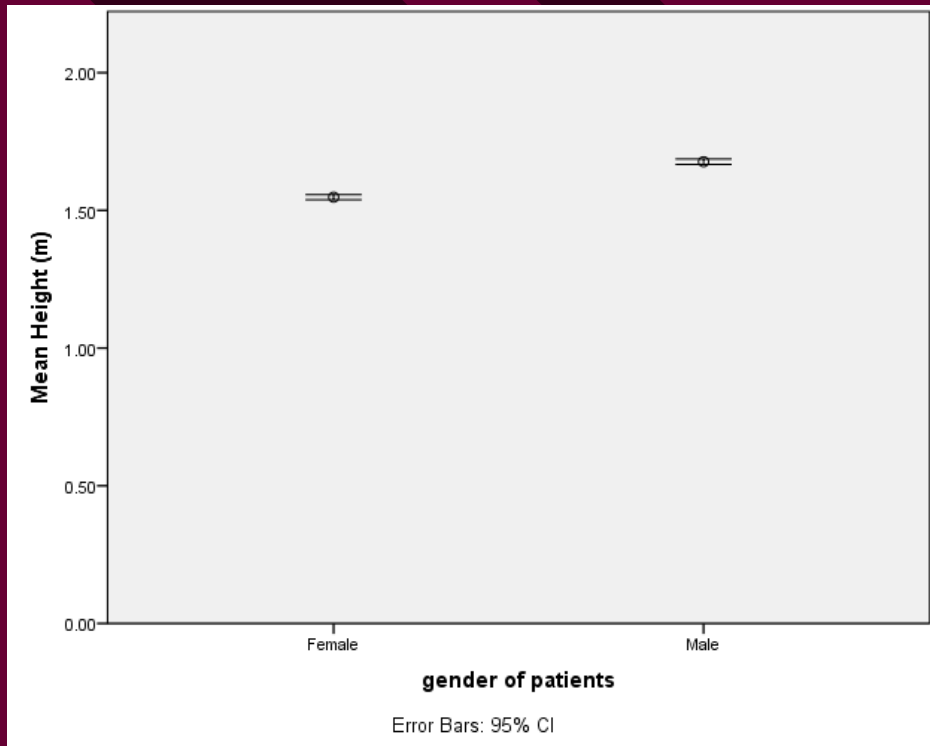
Study subjects

- Subject characteristics are important
- Defined by:
 - Inclusion criteria
 - Exclusion criteria
- Affect generalisability of the conclusions

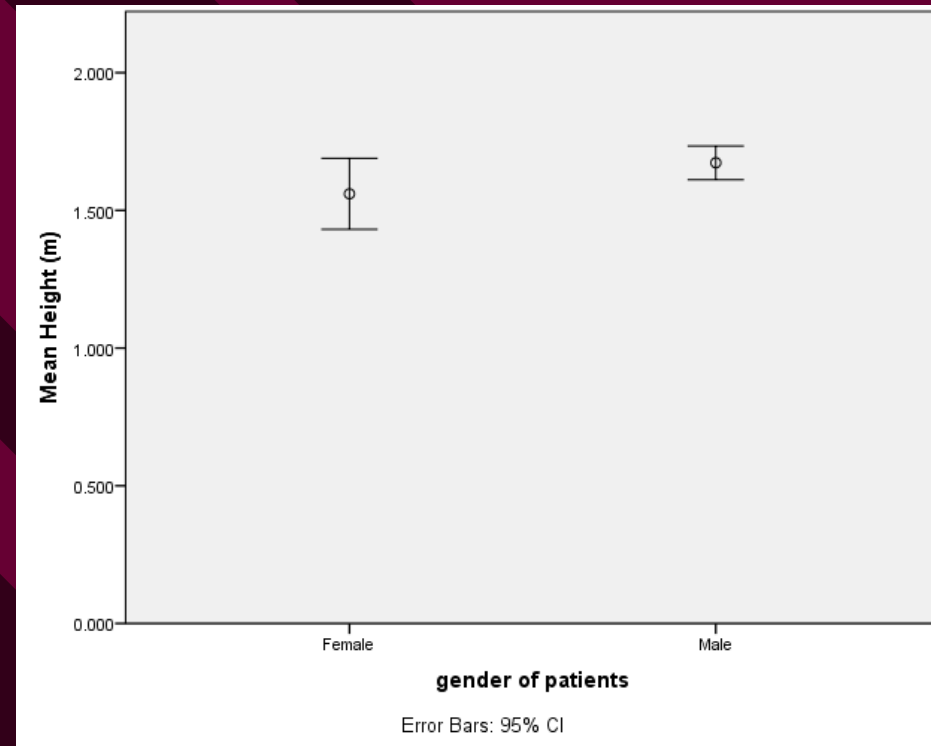
Power of the study

- Usually calculated by a statistician, who needs to know
 - the P value required (usually <0.05)
 - the anticipated difference between treatments relative to the SD (effect size)
 - drop out rate
- Usually at least a power of 80% to detect a difference between treatments at the 5% level of significance is expected in a clinical trial

Sample size and 95% confidence interval



$N = 500$



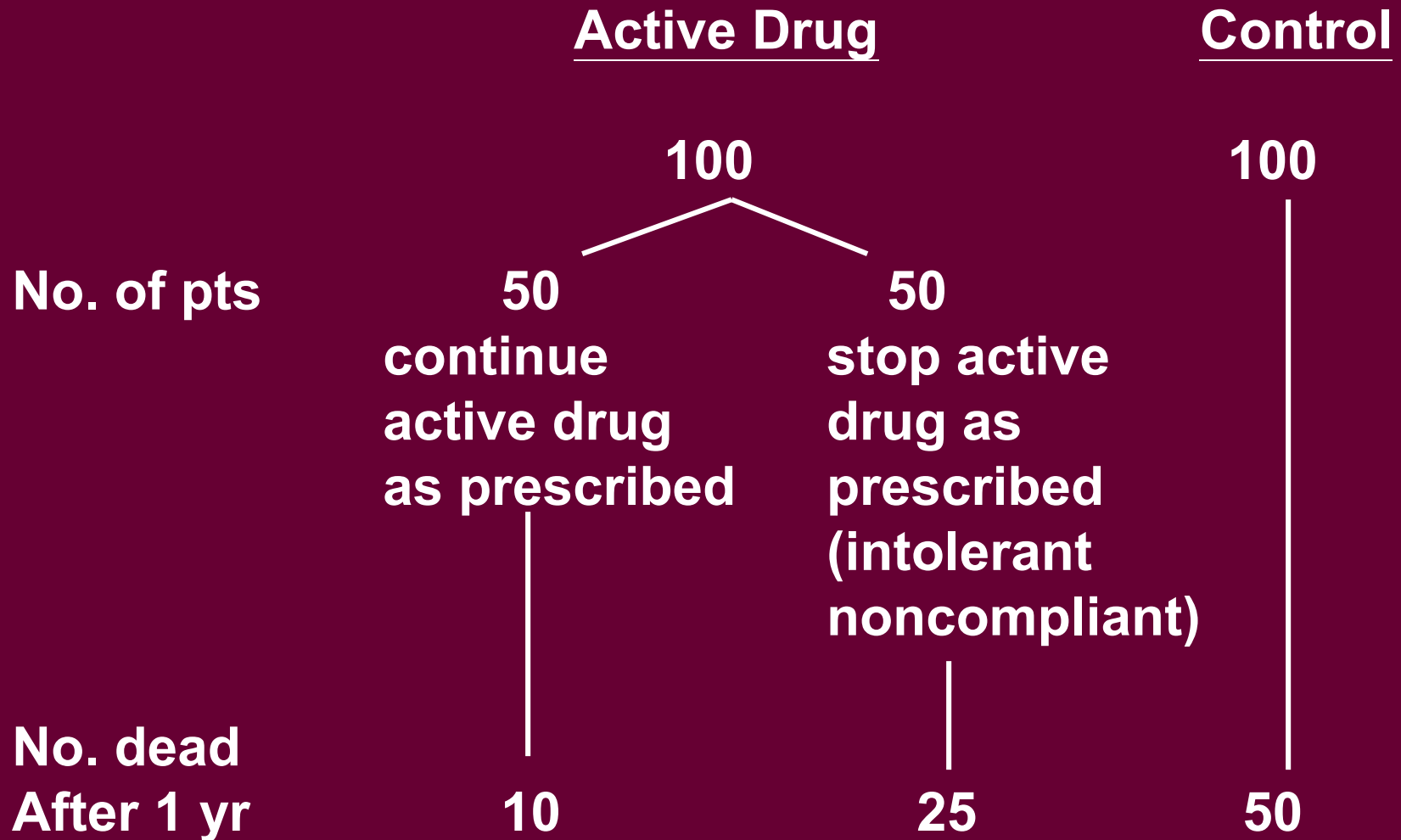
$N = 16$

Adverse events

- Definition – any untoward event (does not need to be related to treatment)
- Reporting – all adverse events must be documented
- Management (no action, observation, treatment, hospitalisation, surgery etc.)
- Outcome (fully or partially recovered, persistent, death)



Efficacy and Effectiveness



Efficacy and Effectiveness

Annual Mortality

	Active Drug	Control
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Efficacy analysis (in those taking drug as prescribed) ‘per protocol’	10/50 (20%)	50/100 (50%)
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Effectiveness analysis (in those offered drug) ‘intention-to-treat’	35/100 (35%)	50/100 (50%)
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Essential statistical terms

- Range, median
- Mean, standard deviation (SD), standard error (SE) of the mean, 95%CI
- Null hypothesis, P value
- T-tests, chi-square test
- Correlation coefficient
- Odds ratio
- Relative risk, relative risk reduction
- Absolute risk reduction, NNT

Two equations you need to know

- $\text{Mean} \pm 1.96 \times \text{SE} \rightarrow 95\% \text{CI}$
- $\text{SE} = \text{SD} / \sqrt{N}$

Hypothesis testing

- Null hypothesis (any difference observed is due to chance)
- P values (the probability of obtaining a test statistic at least as extreme as the one observed)
- Type 1 error (null hypothesis wrongly rejected when it is true, i.e. false positive)
- Type 2 error (null hypothesis not rejected despite being false, i.e. false negative)

Statistical tests used in clinical trials

	Summary statistic	Statistic test	Difference tested
Continuous data	Mean SD SE Median	T-test Mann-Whitney	Mean Rank order
Categorical data	Frequencies	Chi-square	Observed vs. expected frequencies
Time to event “survival data”	Median	Log-rank	Observed vs. expected events

If outcome depends on several factors

	Summary statistic	Statistic test	Multivariate method
Continuous data	Mean SD SE Median	T-test Mann-Whitney	Multiple regression None as sample size usually small
Categorical data	Frequencies	Chi-square	Logistic regression
Time to event “survival data”	Median	Log-rank	Cox proportional hazard

Number-needed-to-treat (NNT) for a hypothetical new drug to prevent stroke

	stroke	no stroke	
drug	10	90	100
placebo	20	80	100
	30	170	200

- Absolute risk of stroke in drug-treated group=10%
- Absolute risk of stroke in placebo control group=20%
- Absolute risk reduction=10%
- Relative risk reduction=absolute risk reduction/absolute risk without treatment=50%
- Number-needed-to-treat= $1 / \text{Absolute risk reduction}$ =10
- NNT=10 means that you prevent one stroke for every 10 persons treated

Limitations of clinical trials

- Patient selection
 - Inclusion/exclusion criteria
 - Other known selection biases
- Protocol restrictions – e.g. fewer drug interactions
- Good medical care, frequent follow up
- Good compliance
- Limited duration of trials, some effects, e.g. carcinogenesis, might be long term
- Very expensive
- Address a limited number of hypotheses

Reliability of different study designs

Study design		Reliability
Case report	Retrospective	-
Case control study	Usually retrospective	+
Cohort	Usually prospective	++
Randomised controlled trial	Usually prospective	+++
Meta-analysis	Usually retrospective	++++
N=1 trial	Usually prospective	+++++

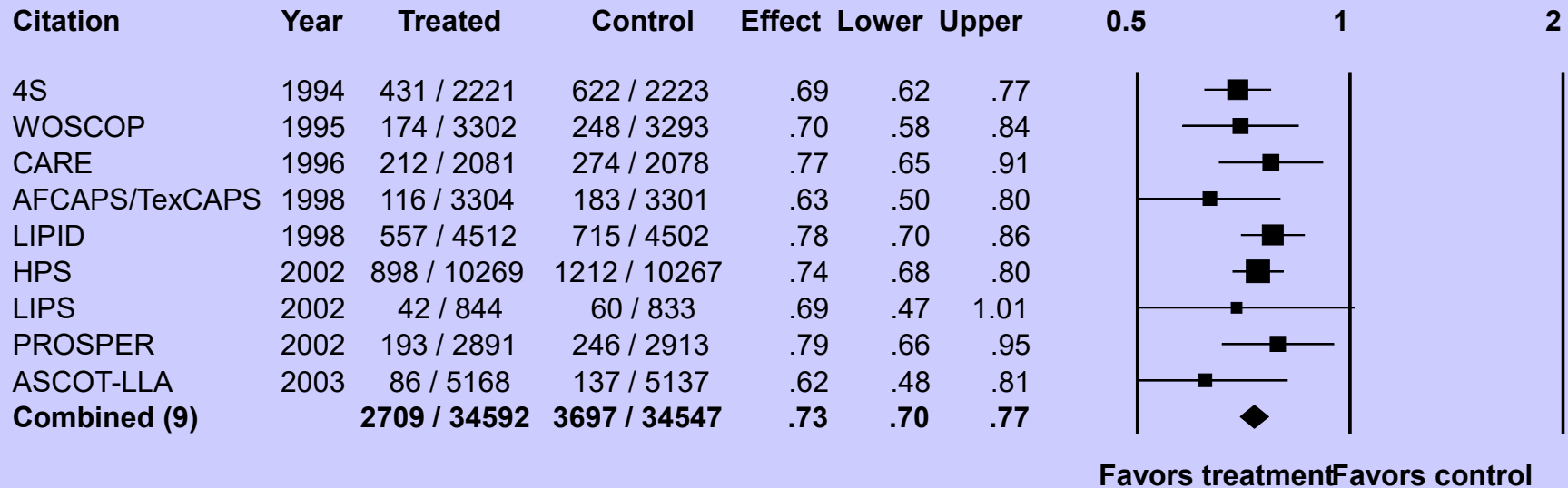
Systematic reviews

- There may be a number of trials addressing the same research question
- The results may vary
- There is a need to summarise the results and arrive at conclusions that take into account of all the trials

Meta-analysis

- A statistical way of combining the results of different studies
- Usually the results of a trial is given a certain statistical weight (a large trial is given more weight)

Major coronary events



Cheung et al. Brit J Clin Pharmacol 2004

Post-marketing surveillance and phase 4 studies

- A drug that has been approved and marketed may still need to be under surveillance
- Rare side effects may not be revealed in clinical trials (if the incidence is 1 in 10,000, then the chance of missing this is 37% when 10,000 patients have been exposed to the treatment)

Post-marketing surveillance

- Clinical trials may not detect long term adverse effects like cancer
- Side effects, if unexpected, may be missed (e.g. cough and ACEIs)
- Standard assessment of adverse effects may miss side effect (valvulopathy after dexfenfluramine)

Reporting adverse drugs reactions

- Voluntary in HK to Drug Office, Dept of Health
- Voluntary in Mainland China, but there are targets
- Yellow card system in the UK
- In USA, reports from professionals, consumers and manufacturers to the FDA Adverse Event Reporting System

Real world studies

Advantages

- Readily extracted from electronic records
- Large numbers possible
- Representative of the real world
- Not highly selected as in clinical trials
- Can study interactions and comorbidities
- Long-term outcomes can be studied

Disadvantages

- Missing information
- Imperfect compliance
- Potential biases

Recommended further reading

- Bland: An introduction to medical statistics
- Pocock: Clinical trials
- Karlberg: Introduction to clinical trials
- Clinical Trials Centre website
<http://www.hku.hk/ctc/index.htm>